

UniVerse

A Campus Companion

Find Teacher

Real-time teacher locator: which room a teacher is in now, time left, and their next class.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Robi (campus explore)

Key files: features/find_teacher/*

1. What it does

Find Teacher answers 'where is this teacher right now?'. You search by name or code and get a card showing their status (In Class / Free / No Class Today), the current room + minutes remaining, and their next class. It is reached from the **Explore** FAB on every dashboard. No new tables - it derives everything from the live `routines` schedule vs the current time.

2. How status is derived

- 1 `fetchAllRoutines()` loads all active routine rows.
- 2 Build a unique teacher list from the rows (code -> name).
- 3 For each teacher, take today's classes sorted by time.
- 4 If `now` is between a class's start and end -> **In Class** (show room, cohort, minutes left = end - now).
- 5 Else if a later class exists today -> **Free** now, show the next class.
- 6 Else -> **No Class Today** (or Free, last class ended).

`\_buildTeacherInfo()` packages this into a `TeacherInfo` object the card renders. Search filters the in-memory list by name or code - instant, no re-query.

3. Real-time

`streamAllRoutines()` is a Supabase Realtime stream on the `routines` table; when the schedule changes the controller rebuilds the teacher list, so locations stay current without a manual refresh.

4. Likely defense questions

question	answer
Is there GPS / live tracking?	No - it's schedule-derived: where the timetable says they should be, compared to the clock. Privacy-friendly and needs no hardware.
Any new backend?	None - it reads the same `routines` table the routine viewer uses.
How is it real-time?	A Supabase Realtime subscription on routines; changes re-derive the list.
What about cancellations?	The feature set reads routines (+ cancellations in the design) to reflect live state; a cancelled class shouldn't show as 'in class'.